



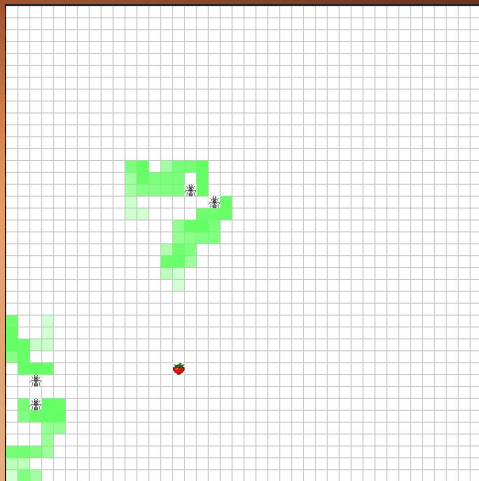
AntHill

Environment

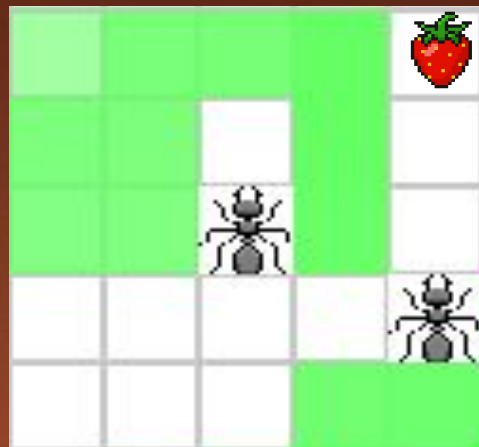
Goals

- Create **training** and **evaluation** frameworks for the environment.
- **Train** and **infer several agents** in parallel in a single environment.
- Explore and discuss **reward functions**

Environment



Agent view - 5x5 grid



Training - Deep Q Network

Neural Network Hyperparameters

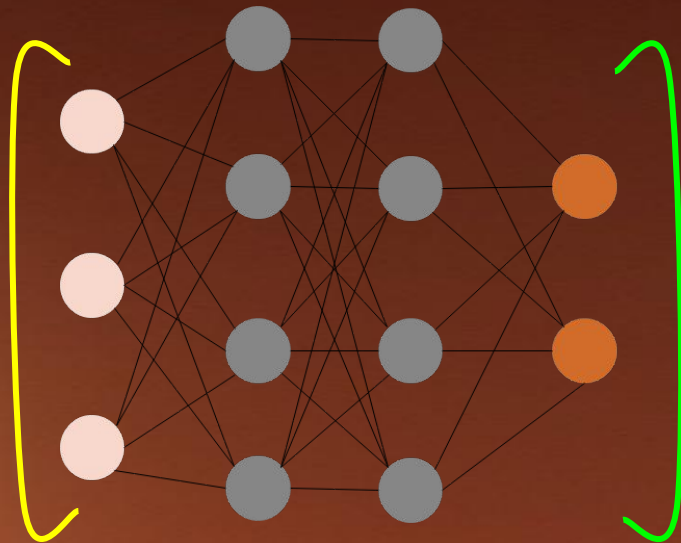
- Learning rate = 0.001
- Activation functions: ReLU (hidden), Argmax(Linear (output))

- Hidden layers: 2×128 units
- Relu

Q-Learning Hyperparameters

- Gamma = 0.99
- Epsilon = 1.0
- Epsilon min = 0.05
- Epsilon decay = 0.995
- Batch size = 64
- Memory size = 20000

Agent observation
space



Agent action
space

Train-evaluate Framework

- Goals **not supported** by either **gymnasium** or **stable baselines**.
- **Model Building**, **Training** and **evaluation** built from scratch.

Classes:



Build Agents
+
Build Models

DeepQLearning
+
Trainer

Evaluator

Execution Framework

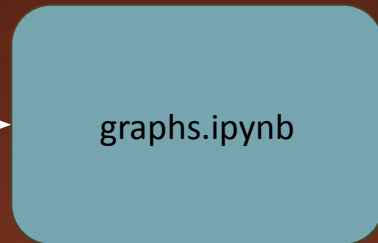
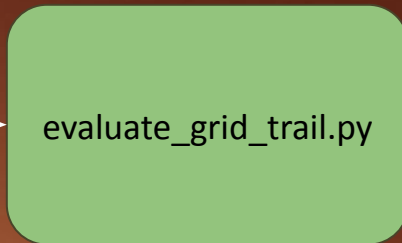
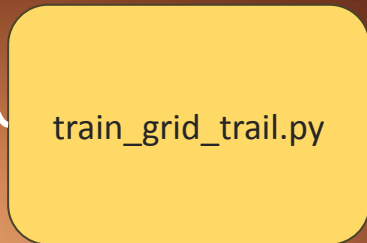
Training, evaluation, plotting

orchestrator.sh

train_grid_trail.py

evaluate_grid_trail.py

graphs.ipynb



Framework - Time

Time considerations:

- 3 reward functions tested
- 1 minute per episode
- 200 training episodes + 40 evaluation episodes per function

=

12 hours per train - evaluate session



Training - V0

Reward function: Global reward pool

+1 if agent found strawberry

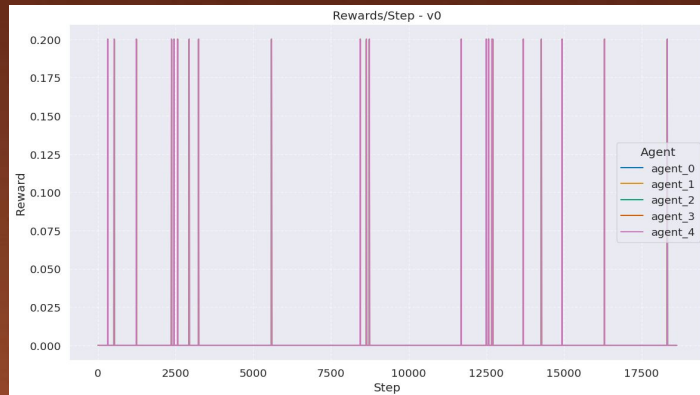
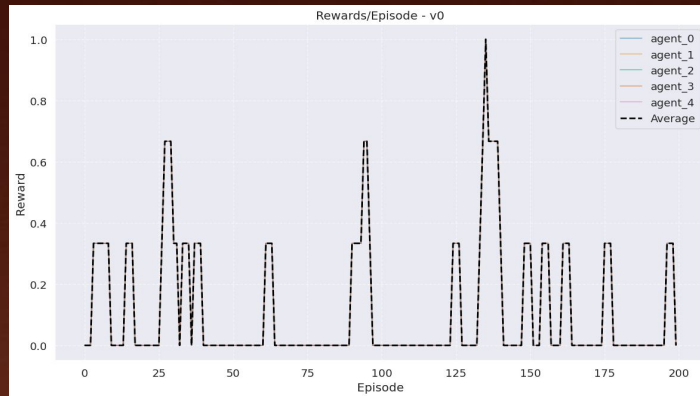
0 otherwise

Goal:

- Encourage emergent behaviors
- Cooperation

Problems

- Reward too sparse
- Not enough episodes for agent to develop complex behaviours



Training - V1

Reward function: Global reward pool

+100 if agent found strawberry

-1 if agent ran over a pheromone cell

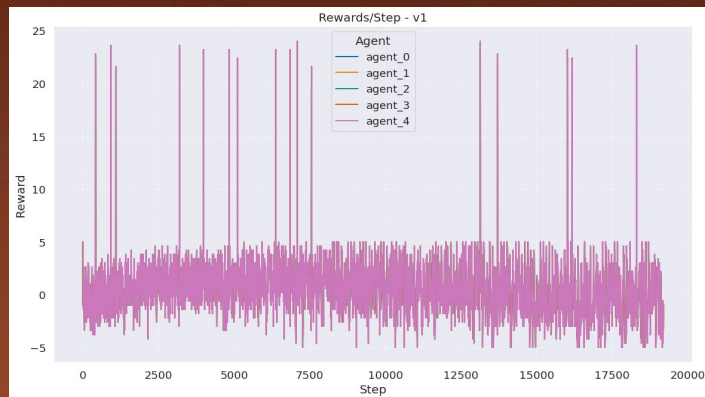
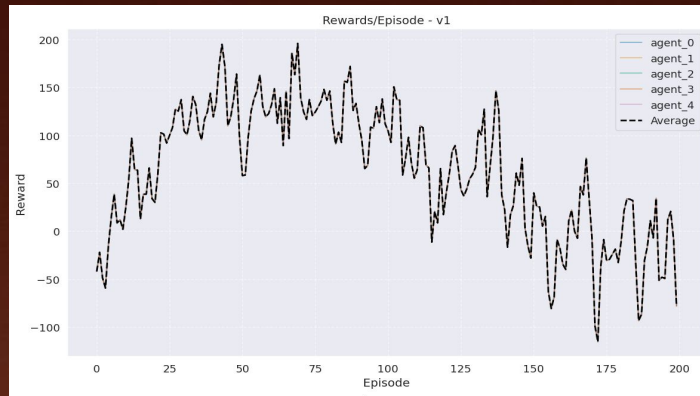
+1 if agent ran over non pheromone cell

Goal:

- Improve **coverage**

Problems

- **Reward pool** introduces **noise**
- No **individualized feedback** for agents
- **Experiences** don't contain **other agents** information



Training - V2

Reward function: Individual Rewards

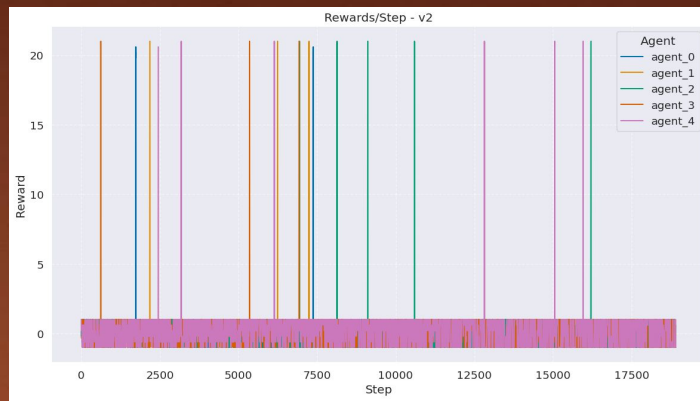
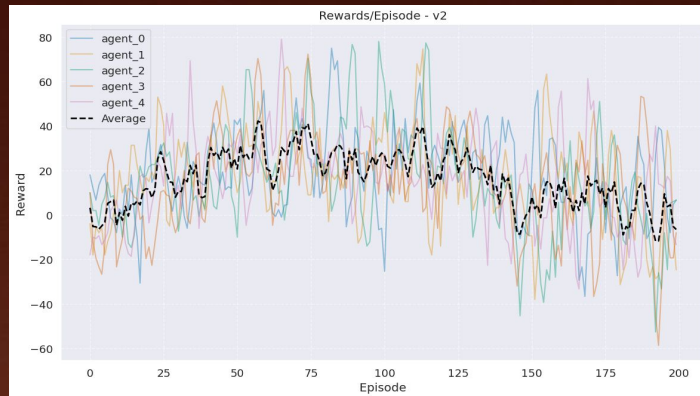
- +100 if agent found strawberry
- 1 if agent ran over a pheromone cell
- +1 if agent ran over non pheromone cell

Goal:

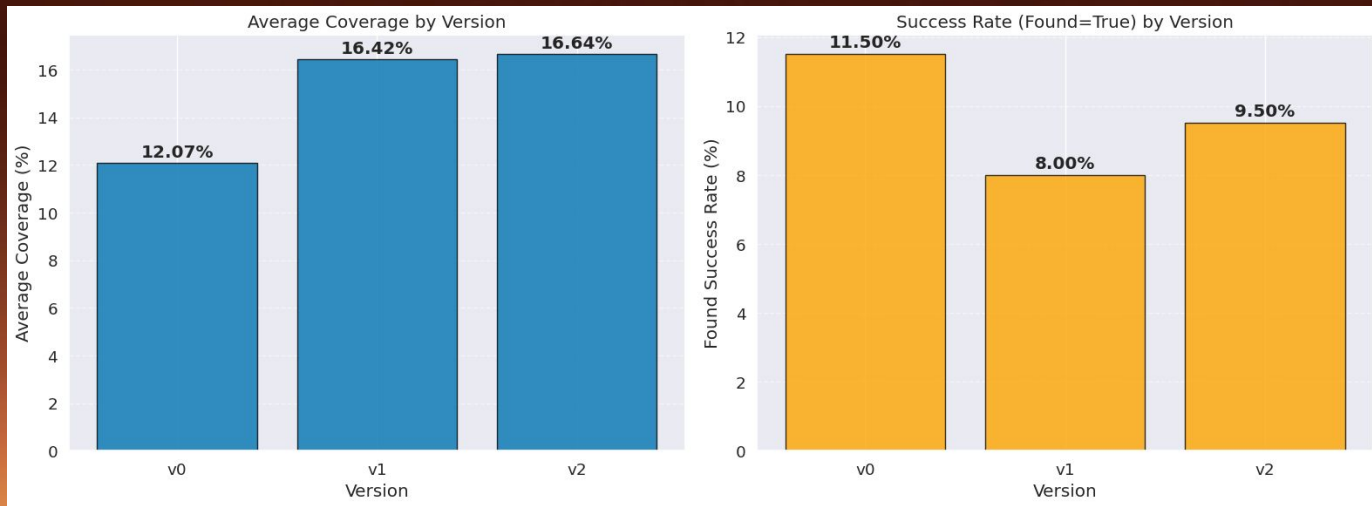
- individualize feedback
- improve learning

Problems

- Reward pool introduces noise
- No individualized feedback for agents
- Experiences don't contain other agents information

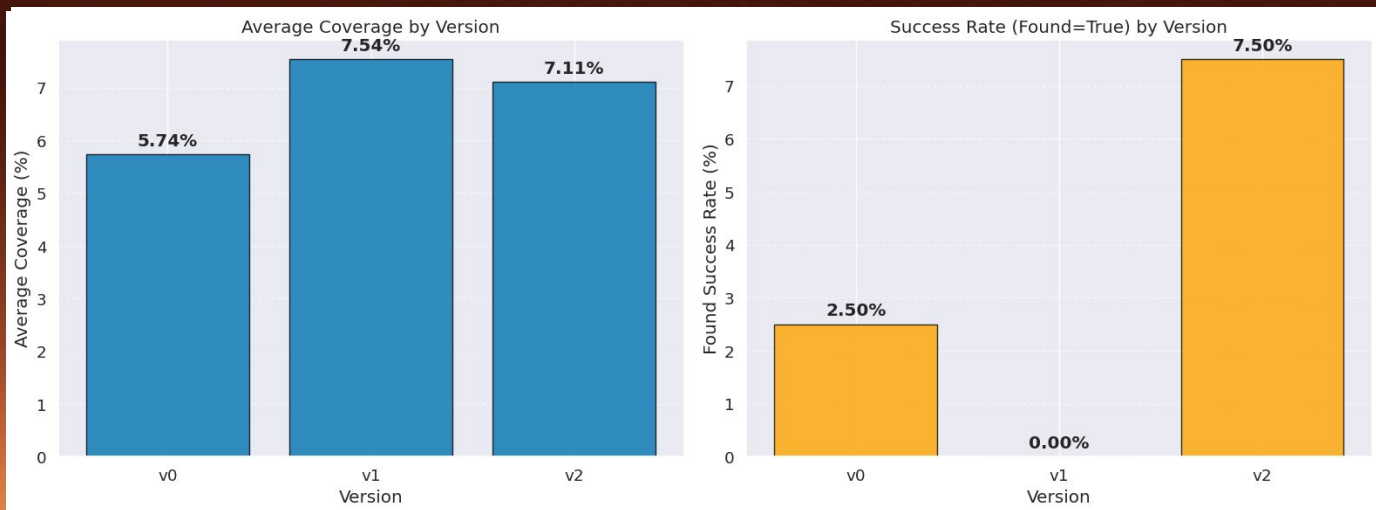


Training Results



- New reward functions **improved** environment **coverage**.
- Agents did not learn to seek the **strawberry**.

Evaluation Results



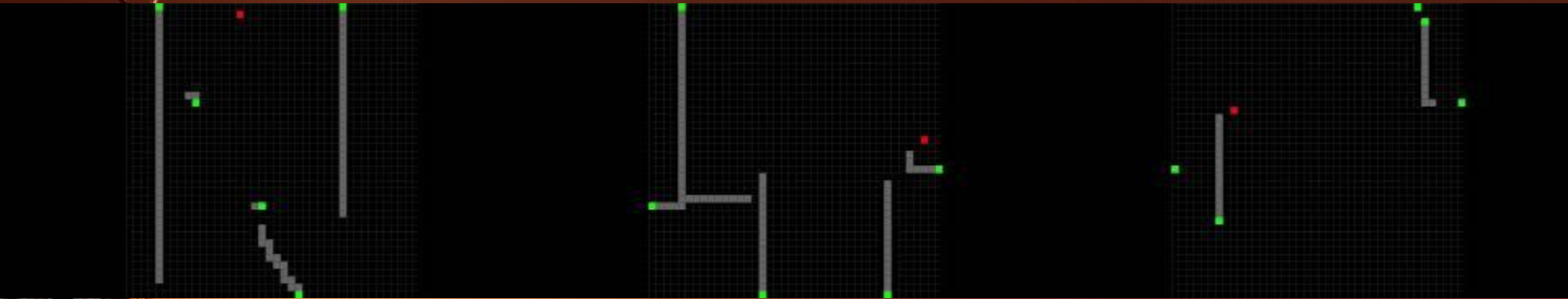
- **V1** and **V2** still present more coverage.
- No trend in **strawberry seeking**.
- Lower coverage due to Epsilon = 0.

Observed behaviours

V0

V1

V2



Next steps

- Curriculum learning → coverage → strawberry
- Switch Learning architecture for a state of the art model (PPO, A2C, etc).
- Fine tune reward function to punish unintended behaviours
- Optimize env for training → migrate to torch

